

Class 9: Candy Mini-Project

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Background

In this mini-project, you will explore FiveThirtyEight's Halloween Candy dataset. FiveThirtyEight, sometimes rendered as just 538, is an American website that focuses mostly on opinion poll analysis, politics, economics, and sports blogging. They recently ran a rather large poll to determine which candy their readers like best.

Importing Candy Data

```
candy_file <- "candy-data.csv"

candy <- read.csv("candy data/candy-data.csv", row.names = 1)

head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1

3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

My favorite candy is Kit Kat and its winpercent is 76.7686

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```
library("skimr")
```

Warning: package 'skimr' was built under R version 4.4.3

```
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, the `winpercent` variable is on a different scale. Most other variables are between 0 and 1, while `winpercent` ranges from 0 to 100.

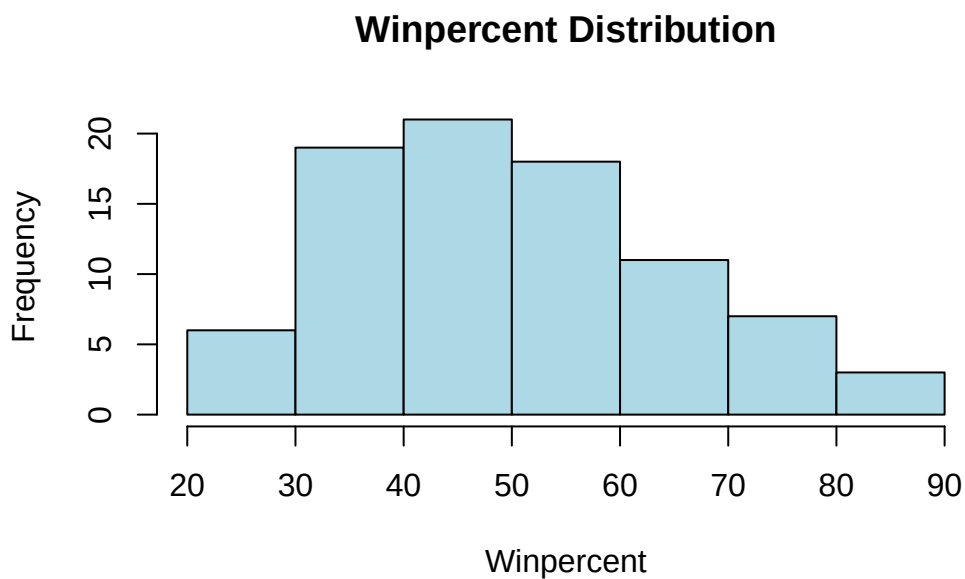
Q7. What do you think a zero and one represent for the `candy$chocolate` column?

A value of 1 means the candy contains chocolate, while 0 means it does not.

Exploratory analysis

Q8. Plot a histogram of `winpercent` values using both base R and `ggplot2`.

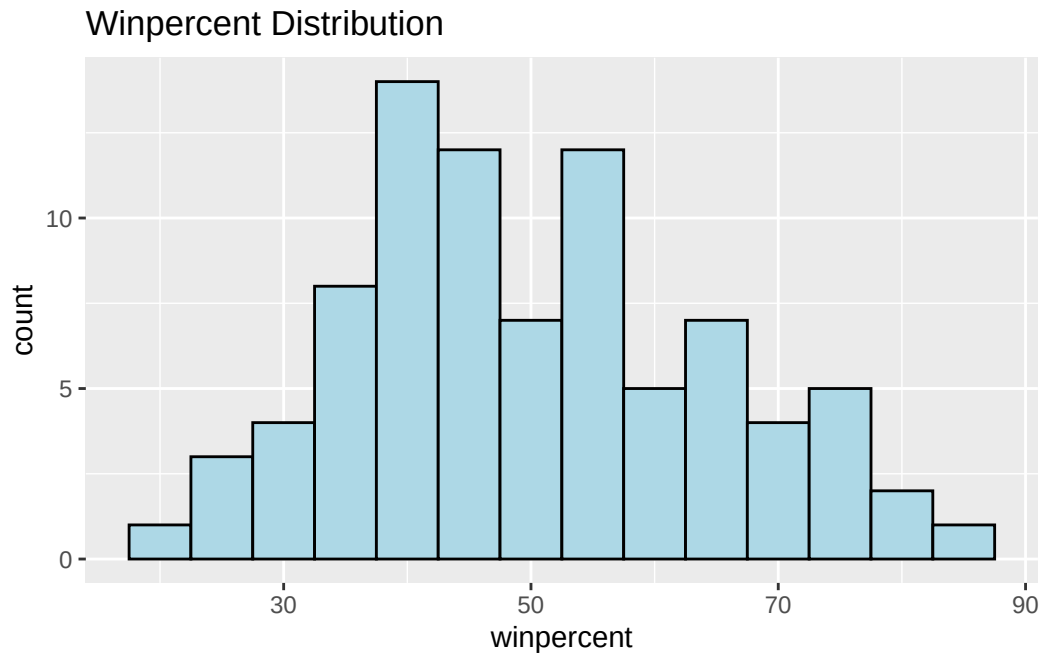
```
hist(candy$winpercent,  
     main = "Winpercent Distribution",  
     xlab = "Winpercent",  
     col = "lightblue")
```



```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
ggplot(candy, aes(x = winpercent)) +  
  geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +  
  labs(title = "Winpercent Distribution")
```



Q9. Is the distribution of `winpercent` values symmetrical?

The distribution of `winpercent` values is not perfectly symmetrical and it is slightly skewed.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

The center of the distribution is slightly above 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

On average, chocolate candies have a higher winpercent than fruity candies.

Q12. Is this difference statistically significant?

```
t.test(
  candy$winpercent[candy$chocolate == 1],
  candy$winpercent[candy$fruity == 1]
)
```

Welch Two Sample t-test

```
data: candy$winpercent[candy$chocolate == 1] and candy$winpercent[candy$fruity == 1]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes, the difference is statistically significant because the p-value (2.871e-08) is much less than 0.05. This shows that there is a significant difference in **winpercent** between chocolate and fruity candies.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
head(candy[order(candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip			0	0	0	1		0.197		0.976
Boston Baked Beans			0	0	0	1		0.313		0.511
Chiclets			0	0	0	1		0.046		0.325

Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511
	winpercent					
Nik L Nip	22.44534					
Boston Baked Beans	23.41782					
Chiclets	24.52499					
Super Bubble	27.30386					
Jawbusters	28.12744					

The 5 least liked candy types are: Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(-candy$winpercent), ], 5)
```

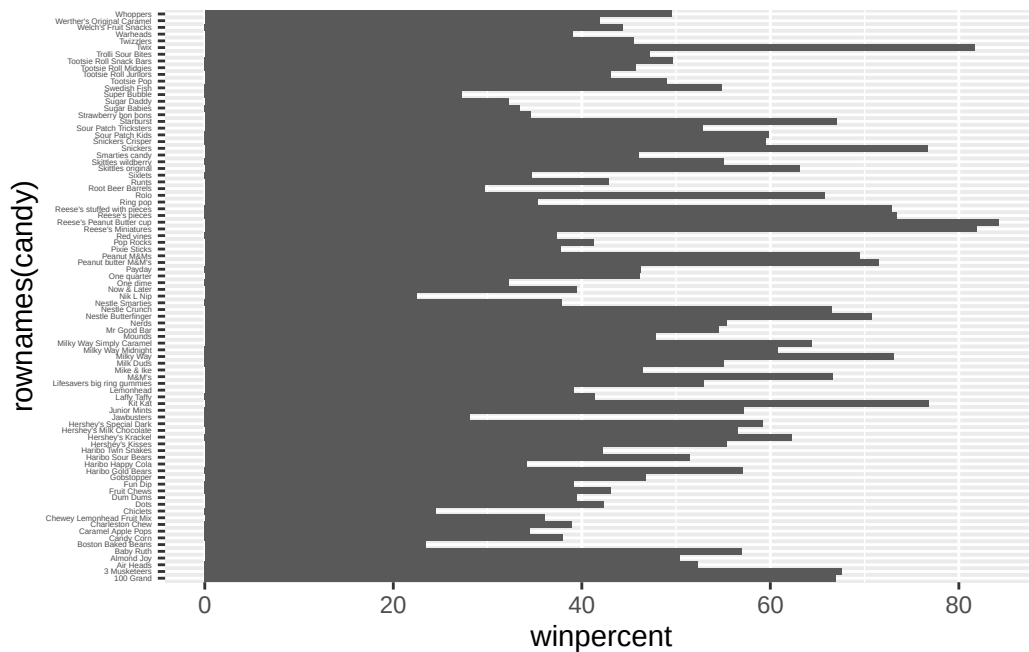
	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1
	crisp	edrice	wafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup		0	0	0	0	0.720
Reese's Miniatures		0	0	0	0	0.034
Twix		1	0	1	0	0.546
Kit Kat		1	0	1	0	0.313
Snickers		0	0	1	0	0.546
	pricepercent	winpercent				
Reese's Peanut Butter cup	0.651	84.18029				
Reese's Miniatures	0.279	81.86626				
Twix	0.906	81.64291				
Kit Kat	0.511	76.76860				
Snickers	0.651	76.67378				

The top 5 favorite candy types are: Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

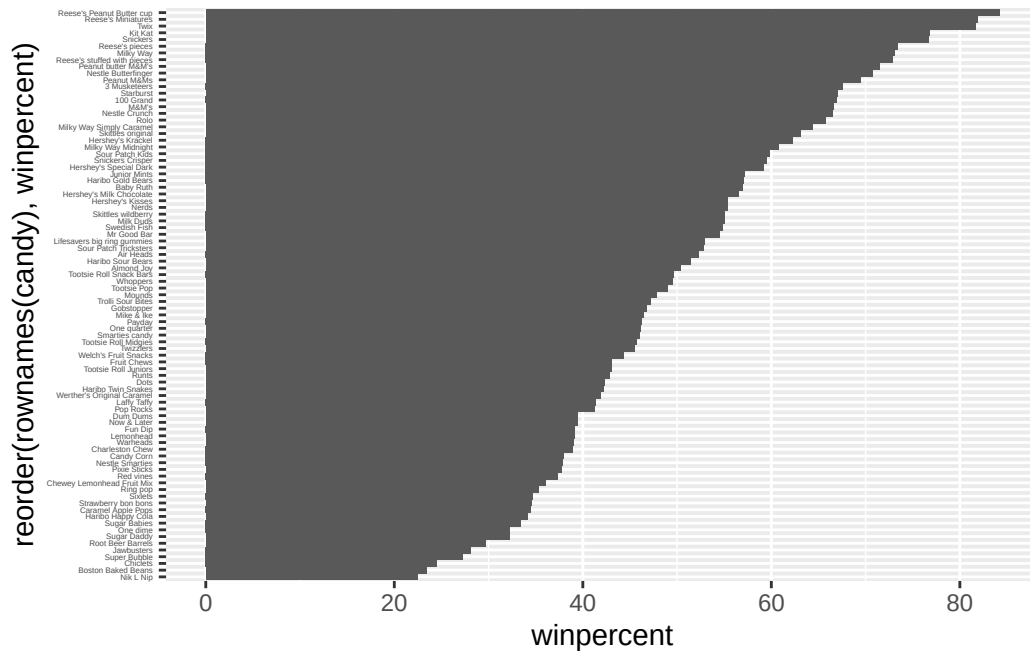

```
library(ggplot2)

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col() +
  theme(axis.text.y = element_text(size = 3))
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col() +
  theme(axis.text.y = element_text(size = 3))
```



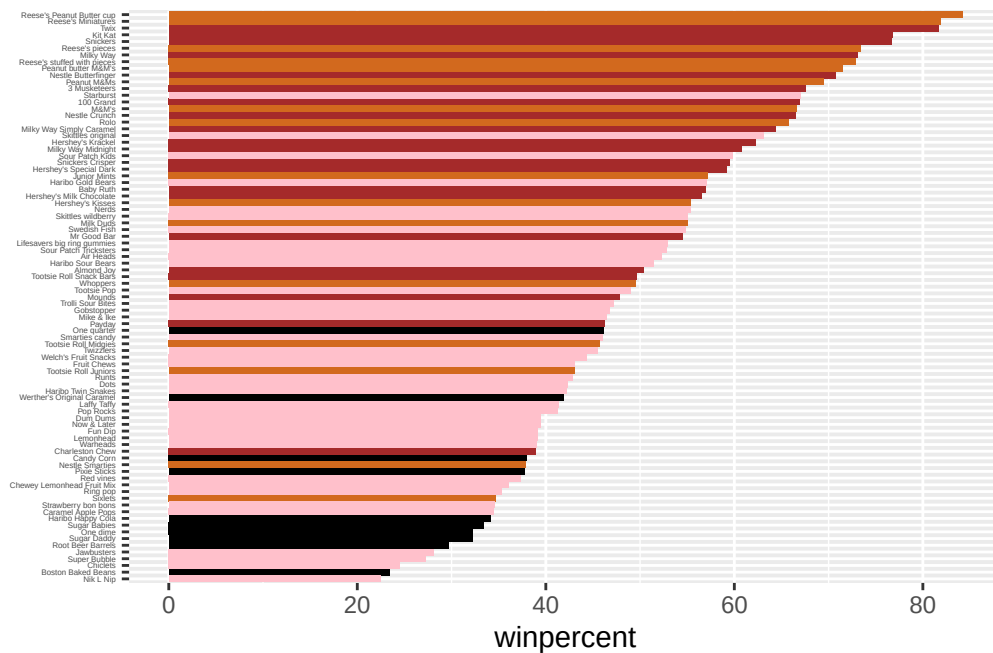
```
my_cols = rep("black", nrow(candy))

my_cols[as.logical(candy$chocolate)] = "chocolate"

my_cols[as.logical(candy$bar)] = "brown"

my_cols[as.logical(candy$fruity)] = "pink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill = my_cols) +
  ylab("") +
  theme(axis.text.y = element_text(size = 3))
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate is Sixlets.

Q18. What is the best ranked fruity candy?

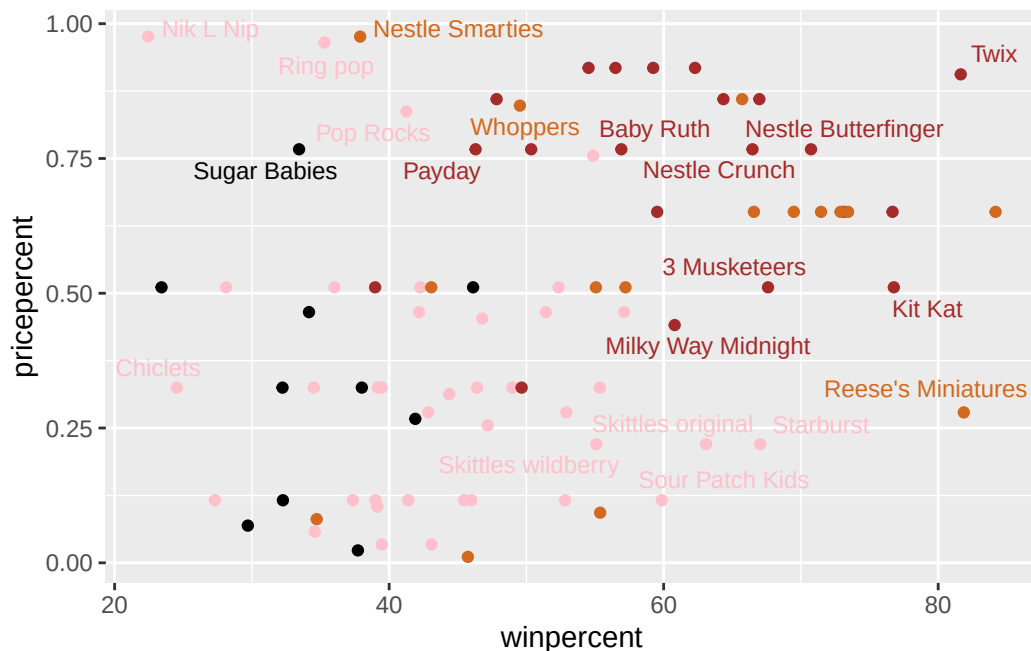
The best ranked fruity candy is Starburst.

Taking a look at pricepercent

```
library(ggrepel)
```

Warning: package 'ggrepel' was built under R version 4.4.3

```
# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label = rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Peanut Butter Cup

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

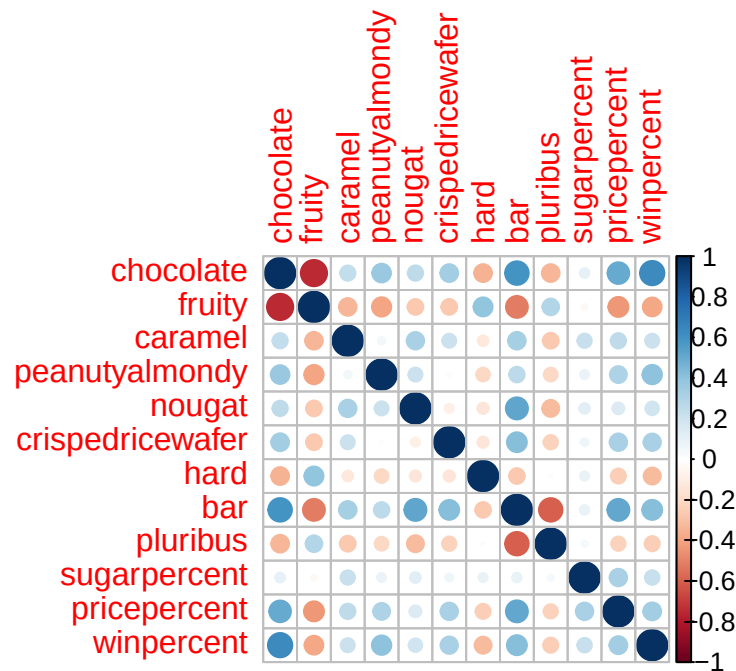
The five most expensive candies are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate. Among these, Nik L Nip is the least popular because it has the lowest winpercent.

Exploring the correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are anti-correlated because candies tend to be either chocolate based or fruit flavored, but not both.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I expect chocolate and fruity to have the largest contributions to PC1 because they are anti-correlated and show a big difference between types of candy, which gives separation along the first principal component.

Principal Component Analysis

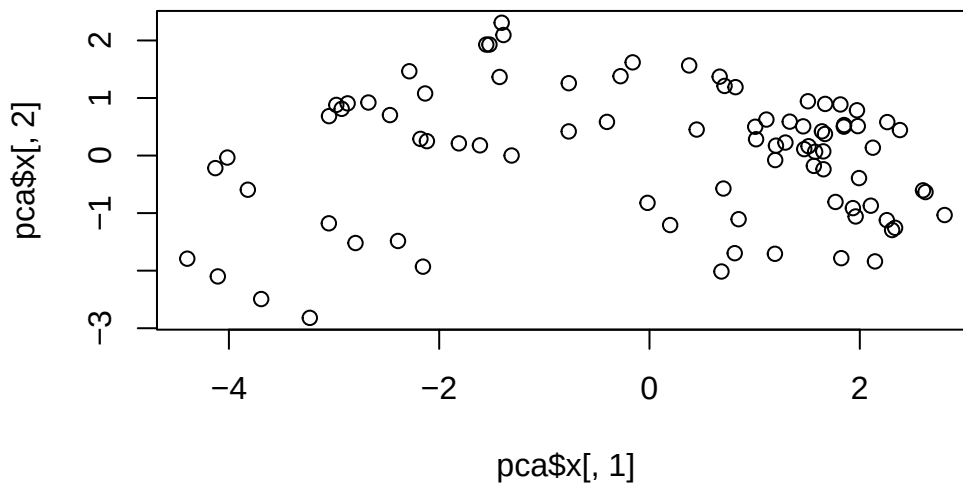
```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

Importance of components:

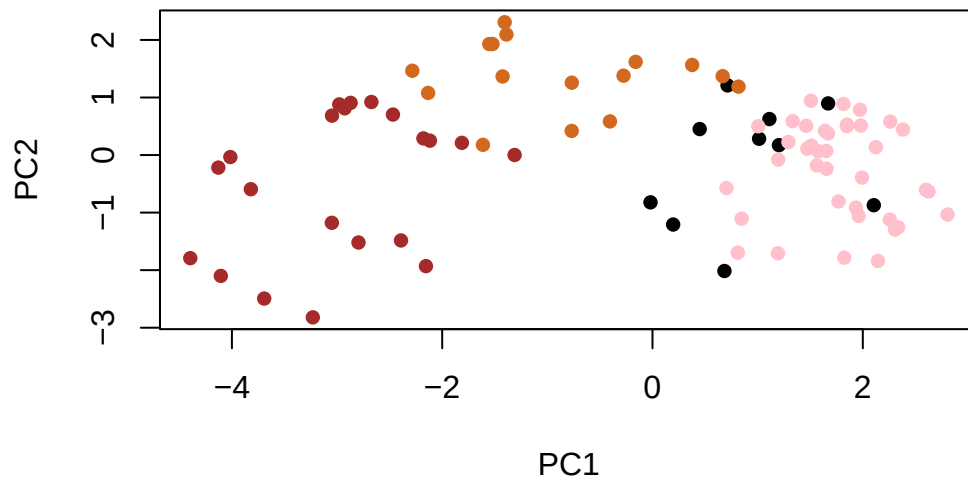
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1], pca$x[,2])
```



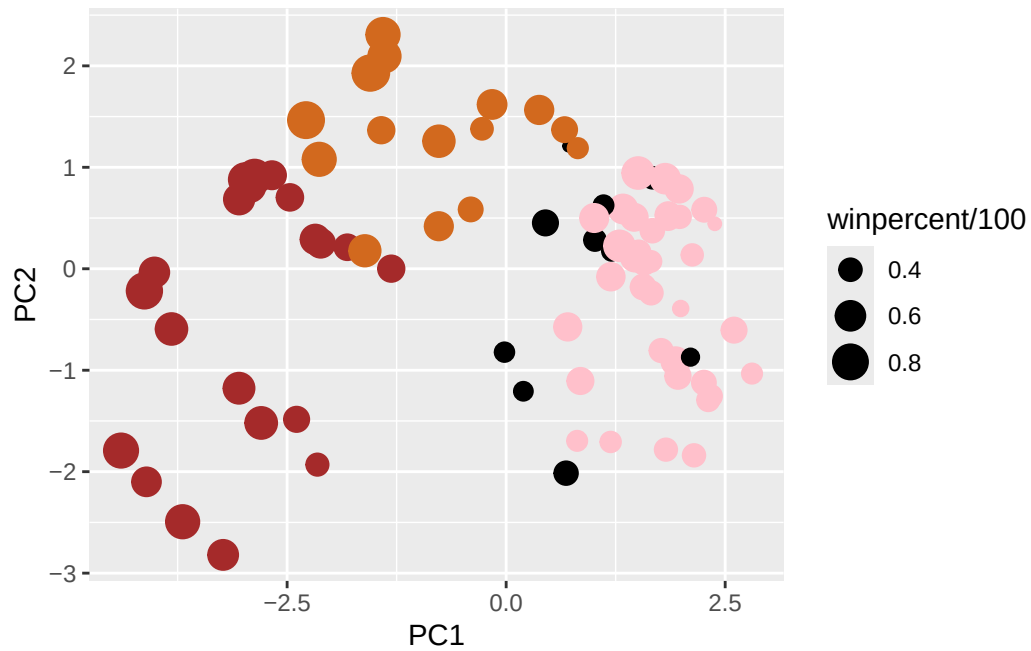
```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +  
  aes(  
    x = PC1,  
    y = PC2,  
    size = winpercent/100,  
    text = rownames(my_data),  
    label = rownames(my_data)  
  ) +  
  geom_point(col = my_cols)
```

p

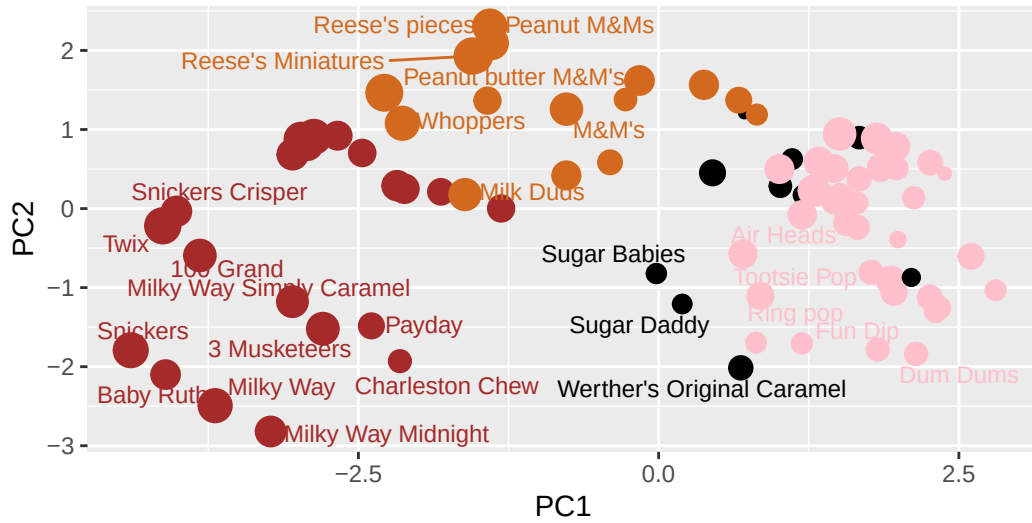


```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

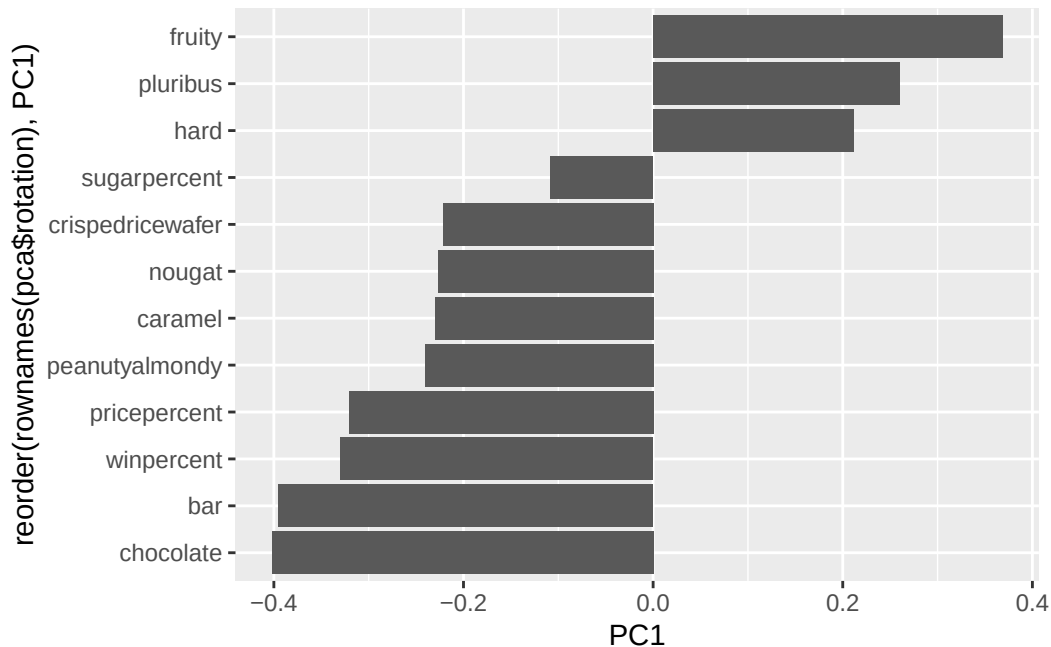

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
ggplot(pca$rotation) +  
  aes(  
    x = PC1,  
    y = reorder(rownames(pca$rotation), PC1)  
  ) +  
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

PC1 has strong positive loadings for fruity, pluribus, and hard, representing fruity multi piece candies. This makes sense and reflects the earlier observed contrast between fruity and chocolate candies.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Winning candies are usually chocolate based, bar type candies with ingredients like caramel or nuts. Visualizations showed higher rankings for these candies, correlation showed chocolate and fruity are opposites, and PCA confirmed this by separating them along PC1. All methods point to the same pattern.